Data Cleaning

Example

TABLE 2.1 Can You Find Any Problems in This Tiny Data Set?

Customer ID	Zip	Gender	Income	Age	Marital Status	Transaction Amount
1001	10048	M	75000	C	M	5000
1002	J2S7K7	F	—40000	40	W	4000
1003	90210		10000000	45	S	7000
1004	6269	M	50000	0	S	1000
1005	55101	F	99999	30	D	3000

CustomerID field is assumed to be fine; But Zip Code, Gender?

- **Zip Code**
  - **Do not assume local format**
    - 90210 (U.S.) vs. J2S7K7 (Canada)
    - In a free trade era should expect some unusual values
  - **Be aware of data type/conversion issues**
    - Zip code 06269 stored in numeric field truncates the leading zeroes, and thus, is represented as 6269 (Zip Code for Storrs, CT)
- **Gender**
  - Value is missing for customer 1003 as shown by arrow.

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- **Income Field Contains $10,000,000?**
  - Possibly valid on zip code 90210 (Beverly Hills, CA)
  - Still considered outlier (extreme data value) - Some statistical and data mining methods negatively affected by outliers.
- **Income Field Contains -$40,000?**
  - Income less than $0?
  - Value beyond bounds for expected income, therefore an error.
  - Caused by data entry error?
  - Discuss anomaly with database administrator.
- **Income Field Contains $99,999?**
  - Value may be valid, but…other values appear rounded to nearest $5,000.
  - Legacy Systems: Value represents database code used to denote missing value?
- **Other considerations for Income**
  - Confirm values in expected unit of measure, such as U.S. dollars.
  - Which unit of measure for income?
  - Customer with zip code J2S7K7 in Canadian dollars?

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- **Age field contains C**
  - Possible a leftover of earlier categorization of age into a bin labeled C?
  - Data Mining software will likely reject a text value on an otherwise numeric field – this needs resolution.
- **Age field contains 0 (zero)**
  - Unlikely: A newborn baby made $1000 transaction.
  - Most probably: Missing value or other anomalous condition coded as 0 (zero).
  - **Important**: Age value will quickly become obsolete; it is recommended to store date type fields (like birthdate) instead, and calculate age as needed.
- **Marital Status Field**
  - What is the meaning of the symbols?
  - Don't make assumptions: Is **S** for Single or Separated?
  - Consider possibility of codes using words from another language: C is for Cold in English, and Chaud (Hot) in French.
- **Transaction Amount Field**
  - Values in this fields seems OK, assuming common unit of measure.